

Apple Inc. (AAPL) Historical Performance Analysis Report

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1. Dataset Overview

The dataset consists of a comprehensive 5-year historical trading window for Apple Inc. (AAPL), sourced directly from Yahoo Finance. The structural framework spans **June 9, 2021, through June 9, 2026**, covering approximately **1,258 trading days** of market data observations.

The core features tracked include:

- **Date:** Organized chronologically as the primary index framework.
- **Open, High, Low, Close:** Daily price parameters measured in USD (\$) capturing intraday ranges and final settlement values.
- **Volume:** A critical market liquidity metric tracking the total number of shares transacted during each individual trading session.

2. Data Transformation Techniques Applied

During the initial data loading phase, a major structural anomaly was identified in the raw source: the CSV file contained a dual-nested metadata header configuration (rows displaying Ticker/AAPL and Date/NaN). Left uncorrected, this forced Pandas to interpret all numerical columns as text strings and improperly labeled the primary timeline axis.

To restore data integrity and prepare the file for mathematical computing, the following advanced transformations were systematically applied:

- **Structural Skipping & Renaming:** Employed `skiprows=3` inside `pd.read_csv()` and injected an explicit schema array (`names=['Date', 'Close', 'High', 'Low', 'Open', 'Volume']`) to cleanly bypass metadata clutter and isolate the true data matrix.
- **Data Type Coercion:** Utilized `pd.to_numeric(..., errors='coerce')` to transform raw text entries into float64 decimals, instantly unlocking analytical computing capabilities.
- **Datetime Alignment & Indexing:** Applied `pd.to_datetime(..., errors='coerce')` to parse string dates into a proper chronological timeline, dropped corrupted rows via `.dropna()`, and permanently bound the dates as the primary row index using `.set_index('Date')`.
- **Imputation of Missing Nodes:** Executed a forward-fill (`.ffill()`) strategy to seamlessly handle non-trading days (weekends and market holidays) by carrying forward the previous session's active closing market price.

3. Features Engineered

To transform raw market metrics into deep behavioral insights, four key features were engineered directly into the data frame:

- **Daily Price Change:** Calculated via `df['Close'] - df['Open']` to measure absolute intraday momentum and net dollar movement within a single market session.

- **Percentage Price Change (Daily Return):** Leveraged `.pct_change()` on closing prices to standardize asset velocity and allow daily rate-of-return performance comparisons.
- **Rolling Momentum Indicators (7-Day & 30-Day Moving Averages):** Utilized `.rolling(window=X).mean()` to filter out daily market "noise" and clearly visualize micro versus macro directional trends.
- **Volatility Measures:** Calculated using a 30-day rolling standard deviation of the daily percentage returns (`.rolling(window=30).std()`) to quantify the asset's active risk premium over time.

4. Trends and Key Insights Identified

- **Macro Price Expansion (2021–2026):** Over the complete 5-year macro timeline, Apple Inc. demonstrates a strong, resilient bullish trajectory. By isolating the full 2021–2026 market boundaries, the stock establishes a definitive historical floor price in the lower ranges of 2021/2022, climbing consistently toward its peak valuation ceilings achieved by mid-2026.
- **Momentum Profiles (Moving Average Crossovers):** The engineered moving averages reveal distinct behavioral characteristics. The **7-Day Moving Average** closely mimics spot market action, reacting instantly to short-term news cycles and earnings data. The **30-Day Moving Average** provides a smoother, stabilized macro baseline. When the 7-Day baseline crosses cleanly above or below the 30-Day baseline, it serves as an early indicator of major shifts in structural market momentum.
- **Liquidity Clusters (Volume Trends):** Market participation is highly clustered rather than uniform. Shaded area visualizations reveal massive, isolated trading volume explosions cutting across the 2021–2026 timeline. These liquidity anomalies align tightly with corporate catalysts, specifically Apple's quarterly earnings calls and annual product keynotes, indicating periods of heavy institutional portfolio rebalancing.
- **Risk Distribution Geometry:** The distribution of daily percentage returns forms a highly symmetrical, tight bell curve centered near 0%. While the asset maintains distinct outlier capabilities—experiencing extreme single-day gains and sharp single-day contractions during periods of broader macroeconomic shifts—the overwhelming majority of trading sessions over the past 5 years reside securely within a low-variance stability zone.

5. Recommendations Based on Findings

- **Institutional Catalyst Tracking:** Because trading volume and rolling volatility expand dramatically around scheduled corporate events, risk models should automatically tighten stop-loss thresholds or temporarily trim exposure in the 48 hours leading up to Apple product keynotes and earnings announcements.
- **Trend-Following Entry & Exit Triggers:** Systematic traders can successfully deploy the 7-Day and 30-Day moving average crossovers mapped out in this analysis. Utilizing these indicators as structural trading signals effectively filters out temporary intraday volatility while capturing large, multi-week market trends.
- **Asymmetrical Risk Budgeting:** Portfolio managers should carefully evaluate maximum single-day contractions versus historical upside spikes when calculating Value at Risk (VaR) targets. Incorporating these specific 5-year extreme volatility parameters ensures the asset is budgeted accurately within broader, multi-asset institutional portfolios.